

7 (a) A computer can perform logical and arithmetic shifts.

(i) Show the result of a logical left shift of 2 places on the two's complement binary integer 11001010

.....
 [1]

(ii) Show the result of an arithmetic right shift of 3 places on the two's complement binary integer 10011110

.....
 [1]

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r}
 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\
 + \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \\
 \hline
 \end{array}$$

[1]

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

